



SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 2241003

GROUP NO: 2

PROJECT TITLE: Explainable Causal ST-GNN Digital Twin for Real-time Air Pollution Forecasting

PROJECT ABSTRACT:

Air Pollution has become a growing problem in urban areas due to rapid city expansions and increased automobile emissions, affecting both human health and our environment. At the same time, current forecasting systems can estimate short-term Air Quality Index (AQI) levels, most of which function as black-box predictive models that do not elucidate the underlying causes of pollution spikes or facilitate policy-level intervention analysis. This project introduces an Explainable Causal Spatio-Temporal Graph Neural Network (ST-GNN) based Digital Twin framework for urban air pollution forecasting and source attribution. Our proposed system models the city as a dynamic graph and its monitoring stations as nodes. The spatial relationships are defined by their geographic proximities and wind-influenced connectivity. A ST-GNN inculcates both spatial interactions among regions and the temporal evolution of pollutants based on their concentrations, supporting accurate multi-horizon AQI prediction. To better understand how the model arrives at its predictions, an attention mechanism is incorporated to estimate the relative influence of traffic-related factors and meteorological conditions on pollution levels. Beyond correlation-based prediction, a causal discovery module is integrated to uncover cause-and-effect relationships among environmental variables. This enables counterfactual simulations, allowing the system to estimate potential AQI variations under hypothetical interventions, such as traffic reduction or emission control measures. To improve practical reliability of the model, prediction uncertainty will also be analysed so that forecast ranges, rather than single-point estimates, can be reported. The system will be validated on real-world air quality and weather datasets and benchmarked against established time series and neural network approaches. Ultimately, the study seeks to design a transparent and policy-relevant forecasting framework that can assist city authorities in understanding, anticipating, and managing pollution trends more effectively.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

A] Software Requirements :

Programming Language: Python 3.9 or above
Deep Learning Framework: PyTorch
Graph Neural Network Library: PyTorch Geometric
Data Processing & Analysis: NumPy, Pandas, Scikit-learn
Statistical Modeling: Statsmodels (for ARIMA and Granger Causality)
Visualization Tools: Matplotlib, Seaborn
Development Environment: Google Colab/ Jupyter Notebook / VS Code
Operating System: Windows / Linux

B] Hardware Requirements (Minimum) :

Processor: Intel i5 (or equivalent)
RAM: 8–16 GB
GPU: NVIDIA GPU with minimum 4–6 GB VRAM (recommended for faster training)
Storage: Minimum 50 GB free disk space
(The project can run on CPU; however, GPU acceleration is recommended for efficient model training.)

C] Methods / Algorithms:

The proposed system will implement a spatio-temporal graph-based learning framework for pollution forecasting.

i) Data Preprocessing: Missing value handling using interpolation, Feature normalization (StandardScaler), Sliding window time-series generation

ii) Graph Construction:

Monitoring stations represented as graph nodes.
Edges formed using geographic proximity (k-nearest neighbors approach)
Adjacency matrix generation

iii) Forecasting Model

Graph Attention Network (GAT) for spatial dependency learning
GRU/LSTM layer for temporal sequence modelling
Fully connected layer for multi-horizon AQI prediction

iv) Explainability

Extraction of attention weights
Feature contribution analysis (traffic and meteorological factors)

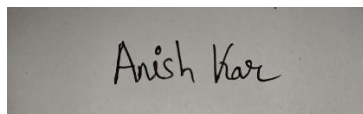
v) Causal Analysis : Granger Causality testing to identify cause-effect relationships

vi) Uncertainty Estimation : Monte Carlo Dropout for prediction interval estimation

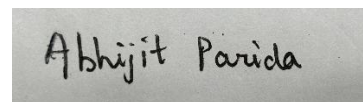
vii) Performance Evaluation : MAE, RMSE, and R^2 metrics, Comparative evaluation with ARIMA and LSTM baselines

(2) NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:

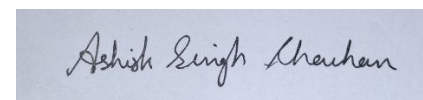
1) Anish Kar 2241011013



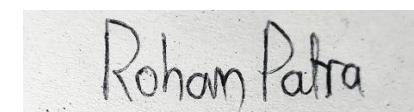
2) Abhijit Parida 2241016184



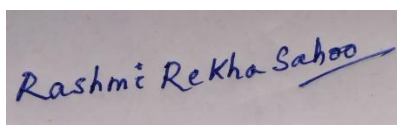
3) Ashish Singh Chauhan 2241003022



4) Rohan Patra 2241016282



(3) APPROVAL STATUS(To be filled in by the Section Coordinator of FRP):



Project Supervisor

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Section Coordinator, FRP

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FRP Coordinator